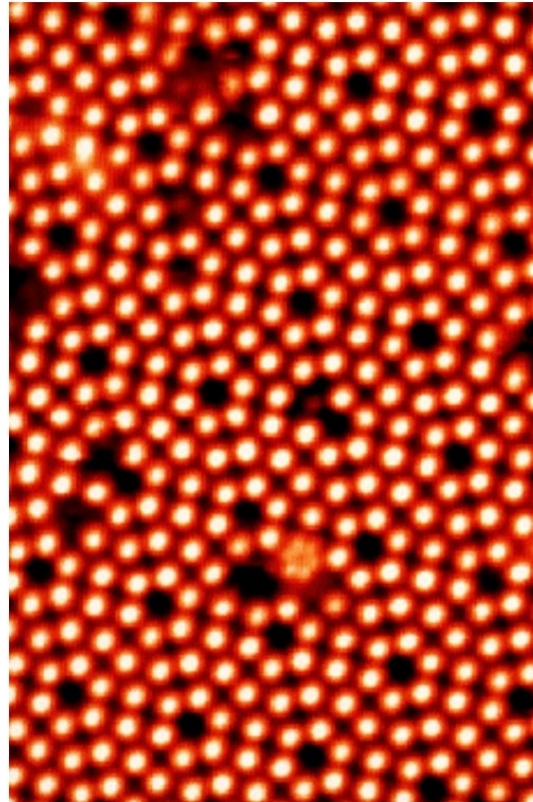


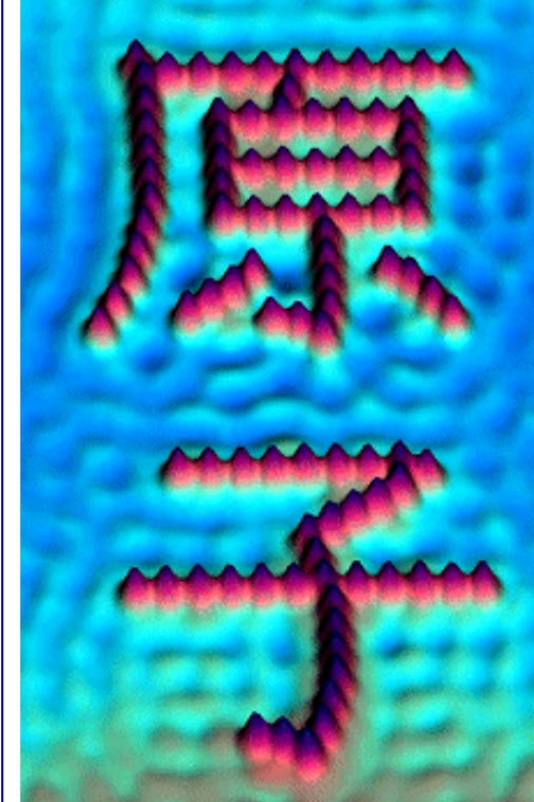
Atoms under the electron microscope



**Scanning Tunneling Microscope
(STM) picture of a silicon surface.**

The surface of a Si crystal as seen with an STM (in ultra-high vacuum). The red-white circles are single silicon atoms. Note that there are *vacancies*, too.

Source: Omnicron Corp.



Drawing with the STM

Si atoms (the red cones) on some "featureless" blue substrate. They have been individually placed to form the Japanese sign for "atom"

Source: IBM STM image gallery

Does atoms have internal structure

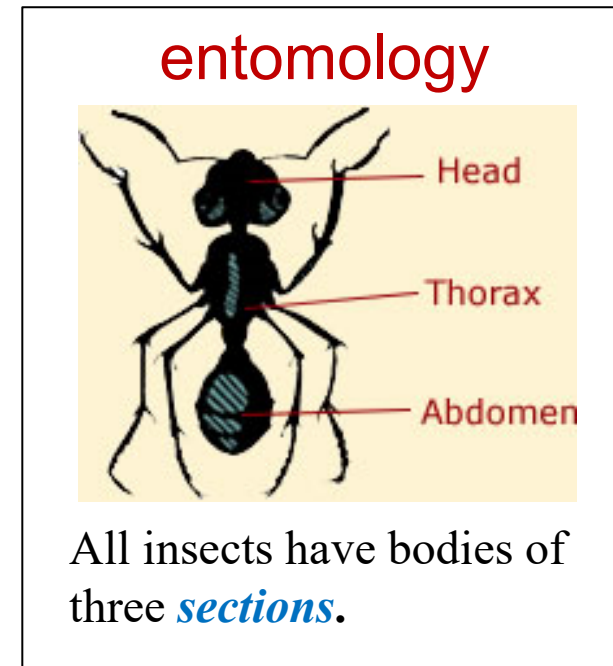
- **atom** (*indivisible, can not be divided*)

a-, an-: not, without

- **a**social Dirac
- **a**chromatic lens
- **a**typical pneumonia
- **a**theists deny God

-tom: a cutting, section

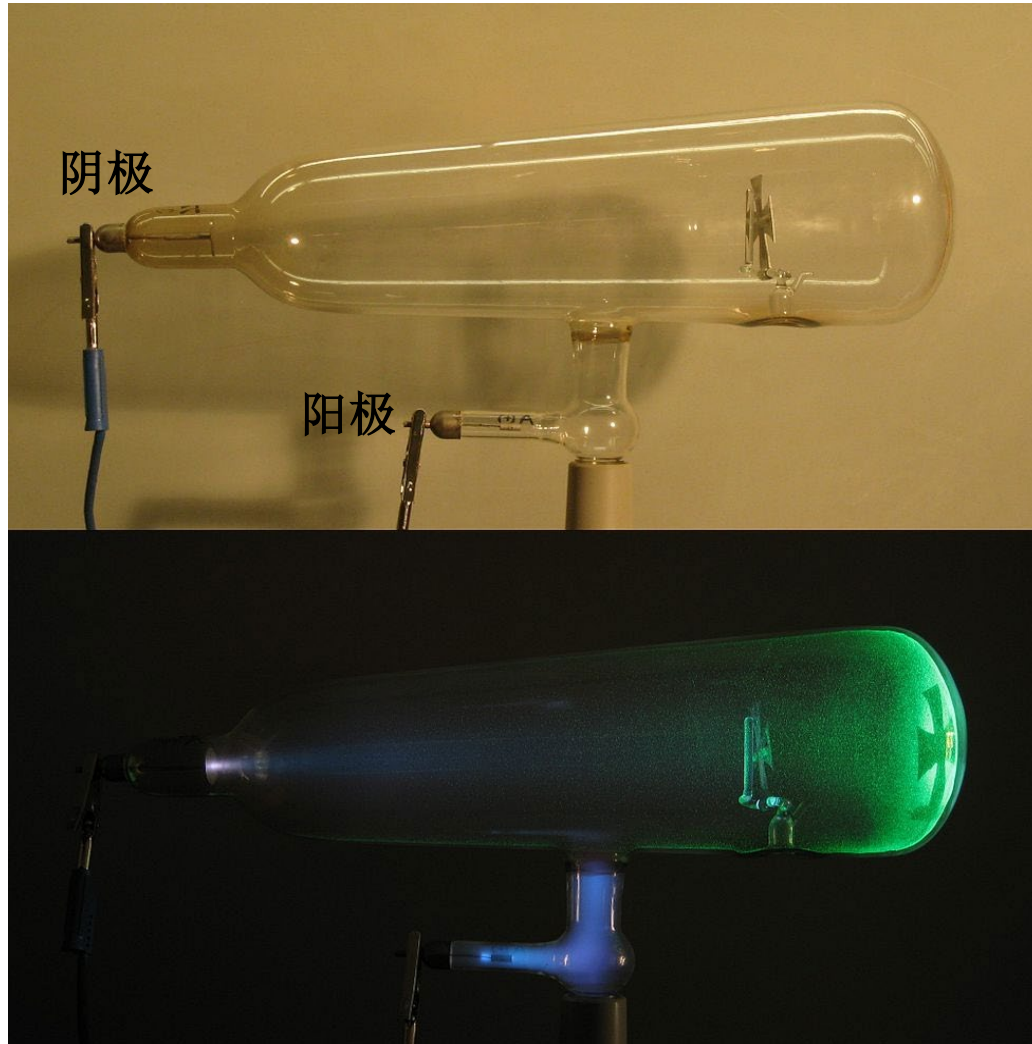
- Ana**tom**y: cut up sth.
- ent**om**ologists love insects
- hystere**ctom**y: womb cutoff



The definitive identification of the electron



Cathode rays



Crookes Tube

Below 10^{-6} atm (atmosphere)

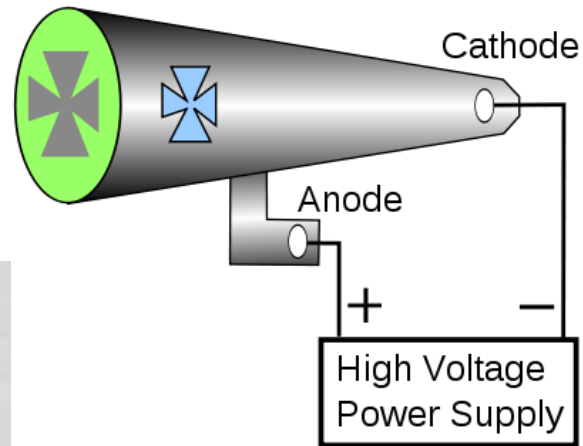
Direct voltage: 10 — 100 kV



Cathode Ray Tube

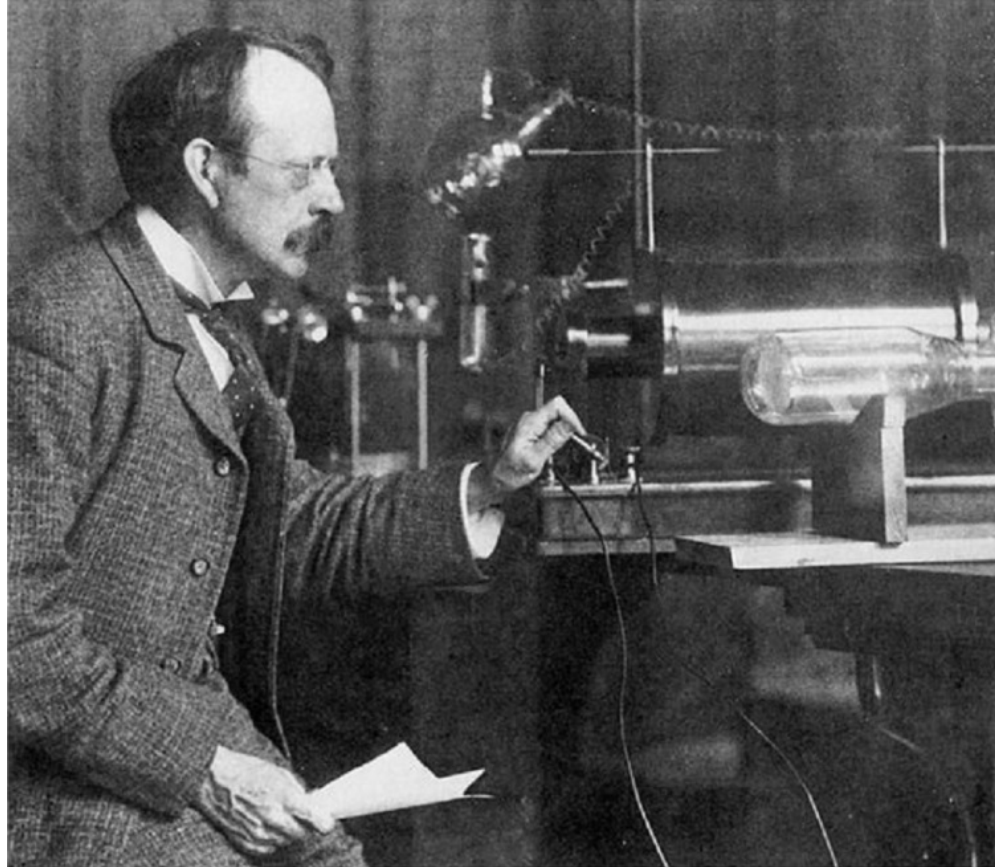
What is nature of cathode rays

Crookes et al. from Britain
A charged particle



Heinrich Hertz et al. from Germany
electromagnetic radiation

J.J Thomson

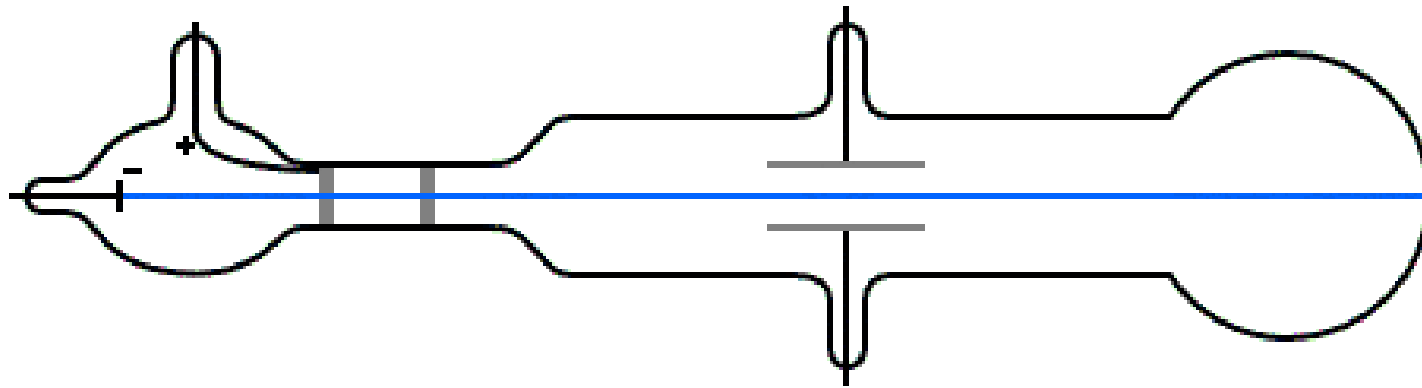


Cavendish laboratory of the University of Cambridge

Intrigued by Rontgen's discovery of X-ray, Thomson set out to study the nature of cathode rays.

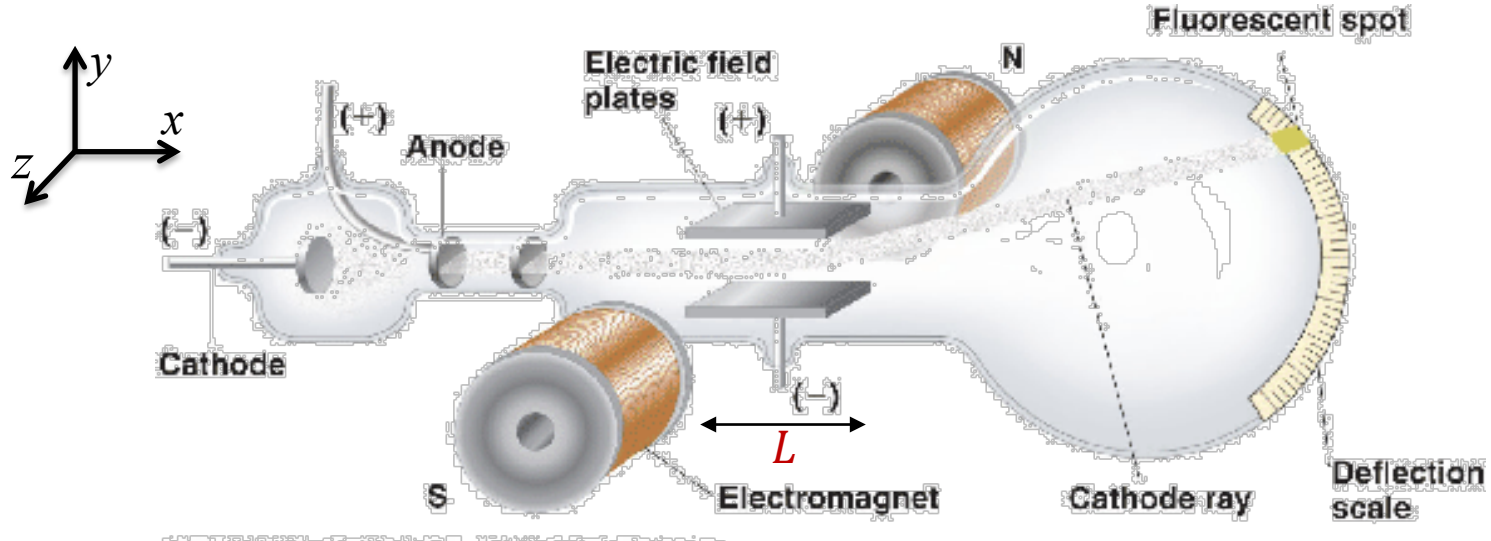
Thomson's cathode ray experiment (1897)

- Cathode ray is negatively charged
- It can be bent by a magnetic or **electric field**



- Its properties are independent of the material of the cathode.
- **Measured its charge-to-mass ratio**

Electron's charge-to-mass ratio



e : electron charge, m : electron mass, L : length of the E-field plate, v : velocity

1. Apply a electric field E along y , bending the ray by a small deflection angle θ .

$$\theta \approx \frac{\Delta P_y}{P_x} = \frac{E e L / v}{m v}$$

2. Then apply a magnetic field B along z to straighten the ray.

$$e v B = e E \rightarrow v = E / B$$

$$\left. \begin{array}{l} \theta \approx \frac{\Delta P_y}{P_x} = \frac{E e L / v}{m v} \\ e v B = e E \rightarrow v = E / B \end{array} \right\} \Rightarrow \frac{e}{m} = \frac{E \theta}{B^2 L}$$

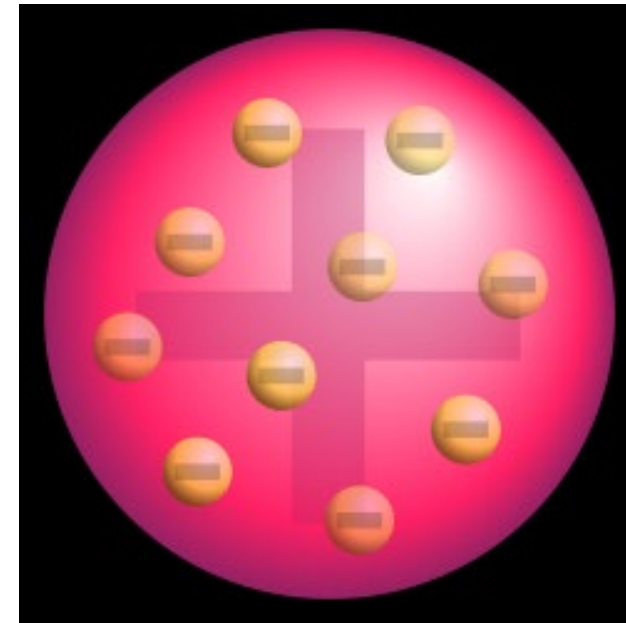


Thomson's achievement

- Definitively identified the electron, the first subatomic particle to be discovered.
- Measured electron's charge-to-mass ratio
 - Through observations in the cloud chamber, he also found the charge of the electron to be comparable to that of the hydrogen ion, thus the mass of an electron is $1/1800$ that of a hydrogen ion.

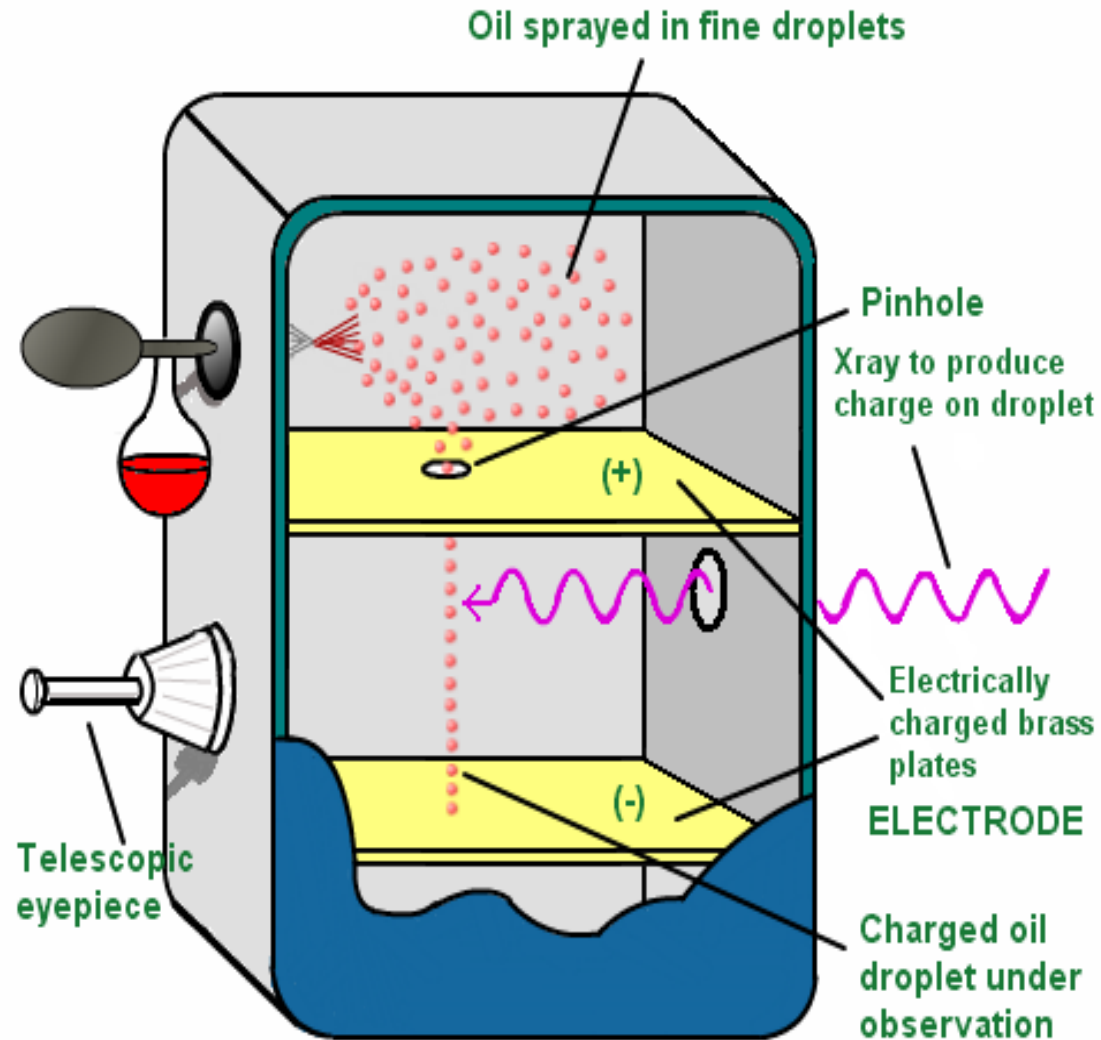
Thomson's plum-pudding model of atoms

Negatively charged electrons are embedded in a uniform positively charged sphere.



The charge of electron

Millikan's oil drop experiment (1910)

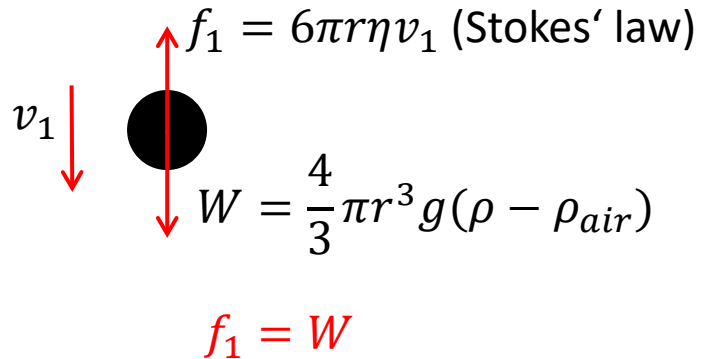


Millikan improved
Thompson's cloud chamber
method



The theoretical model of the oil drop experiment

Step 1: measure the speed of a free fall oil drop


$$f_1 = 6\pi r \eta v_1 \text{ (Stokes' law)}$$
$$W = \frac{4}{3} \pi r^3 g (\rho - \rho_{air})$$
$$f_1 = W$$

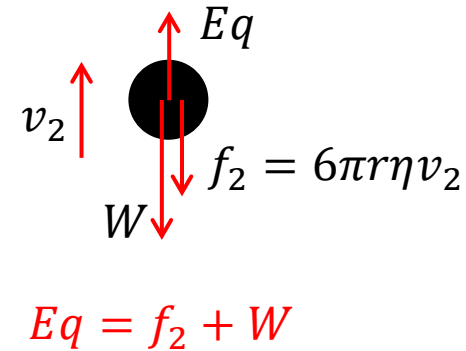
η : viscosity of oil in air, v_1 : free-fall terminal velocity, ρ : oil density, ρ_{air} : air density,

W : drop's apparent weight, f_1 : air friction, r : drop radius

$$f_1 = W \Rightarrow r \Rightarrow W$$

* [Viscosity of a small sphere in air](#)

Step 2: Apply E to push the charged drop upwards


$$Eq = f_2 + W$$

E : E-field strength, v_2 : up-going terminal velocity, q : charge of the drop, f_2 : air friction forces,

Using $\frac{f_2}{f_1} = \frac{v_2}{v_1}$, $f_1 = W$ and $Eq = f_2 + W$, gives

$$q = \frac{v_1 + v_2}{E v_1} W$$

Results: All measured charges were integer multiples of a fundamental charge $e = 1.5924(17) \times 10^{-19} \text{ C}$

Millikan's discovery

- Charge is quantized with a minimum unit
- Electron charge = $1.5924(17) \times 10^{-19} \text{ C}$,

Modern day value: $1.602176487(40) \times 10^{-19} \text{ C}$

Automated Millikan experiment measured 10^8 oil drops in 20 y's

